Speed, Distance & Time

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Easy / Standard

1. 60 km/h (60 ÷ 1 = 60)

2. A — Distance = Speed × Time

3. 60 km (20 × 3 = 60)

4. 200 km/h (400 ÷ 2 = 200)

5. C — Speed combines distance and time: km/h

6. 20 km (5 × 4 = 20)

7. 30 km/h (90 ÷ 3 = 30)

8. 160 km (80 × 2 = 160)

9. A, C and D (100÷2=50, 90÷3=30 and 40÷4=10 are correct; 60÷3 is actually 20 (not 30) and 24÷2 is actually 12 (not 8))

10. 50 km/h (150 ÷ 3 = 50)

11. 4 km/h (8 ÷ 2 = 4)

12. B — 45 km/h is the largest value

13. 100 km (50 × 2 = 100)

14. 50 km/h (200 ÷ 4 = 50)

15. 12 km/h (24 km ÷ 2 hours = 12 km/h)

Standard

16. 60 km/h (240 ÷ 4 = 60)

17. 60 km (15 × 4 = 60)

18. B — 180 ÷ 3 = 60

19. 36 km (12 × 3 = 36)

20. 70 km/h (210 ÷ 3 = 70)

21. 200 km (40 × 5 = 200)

22. C — 600 × 2 = 1,200

23. 15 km/h (45 ÷ 3 = 15)

24. 80 km/h (320 ÷ 4 = 80)

25. 30 km (6 × 5 = 30)

26. A, C and E (50×2=100, 50×1=50 and 50×5=250 are correct; 50×3 is actually 150 (not 200) and 50×4 is actually 200 (not 300))

27. 70 km/h (350 ÷ 5 = 70)

28. 180 km (45 × 4 = 180)

29. B — 100 ÷ 2 = 50 km/h, the highest of the four (40, 50, 45, 30)

30. 50 km/h (300 km ÷ 6 hours = 50 km/h)

Standard / Hard

31. 160 km ÷ 4 hours = 40 km/h

32. B — 180 ÷ 60 = 3

33. 30 km ÷ 3 hours = 10 km/h

34. 3 hours (600 ÷ 200 = 3)

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35. 80 km (40×2=80) + 50 km (50×1=50) = 130 km

36. B — Return takes 4 hours; 180 ÷ 4 = 45

37. 60 km (20×3=60) + 60 km (30×2=60) = 120 km

38. (30+50) × 2 = 160 km

39. A, C, D and E (200÷40=5 hours, so B is false; in 2 hours, 40×2=80km is covered; half the journey, 100km, takes 100÷40=2.5 hours; and at 40 km/h the car always covers 40 km every hour)

40. Each half is 120 km. First half: 120÷60=2 hours. Second half: 120÷40=3 hours. Total = 5 hours.

41. 40÷20=2 hours, +1 hour rest, +20÷10=2 hours = 5 hours total

42. C — 2 hours × 80 km/h = 160 km

43. 240 km ÷ 4 hours = 60 km/h

44. (20−8)×2 = 24 km

45. First part takes 180÷60=3 hours; remaining time = 5−3=2 hours; 120÷2=60 km/h

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Homework Worksheet — Answers

Easy / Standard

1. 40 km/h (40 ÷ 1 = 40)

2. B — Speed = Distance ÷ Time

3. 30 km (15 × 2 = 30)

4. 50 km/h (100 ÷ 2 = 50)

5. B — km/h combines distance (km) and time (hours)

6. 12 km (4 × 3 = 12)

7. 60 km/h (120 ÷ 2 = 60)

8. 120 km (30 × 4 = 120)

9. A, C and D (80÷2=40, 100÷5=20 and 45÷3=15 are correct; 60÷2 is actually 30 (not 20) and 90÷3 is actually 30 (not 20))

10. 60 km/h (180 ÷ 3 = 60)

11. 3 km/h (6 ÷ 2 = 3)

12. B — 20 km/h is the smallest value

13. 110 km (55 × 2 = 110)

14. 70 km/h (210 ÷ 3 = 70)

15. 12 km/h (24 km ÷ 2 hours = 12 km/h)

Standard

16. 60 km/h (180 ÷ 3 = 60)

17. 50 km (10 × 5 = 50)

18. B — 240 ÷ 4 = 60

19. 42 km (14 × 3 = 42)

20. 70 km/h (280 ÷ 4 = 70)

21. 140 km (35 × 4 = 140)

22. C — 500 × 3 = 1,500

23. 15 km/h (60 ÷ 4 = 15)

24. 90 km/h (270 ÷ 3 = 90)

25. 30 km (5 × 6 = 30)

26. A, C and D (60×2=120, 60×1=60 and 60×4=240 are correct; 60×3 is actually 180 (not 200) and 60×5 is actually 300 (not 280))

27. 65 km/h (260 ÷ 4 = 65)

28. 144 km (48 × 3 = 144)

29. D — 210÷3=70 km/h, the highest of the four (50, 60, 40, 70)

30. 60 km/h (300 km ÷ 5 hours = 60 km/h)

Standard / Advanced

31. 120 ÷ 40 = 3 hours = 180 minutes

32. A — 2 hours 30 minutes = 2.5 hours; 100 ÷ 2.5 = 40

33. 30 minutes = 0.5 hours; 400 × 0.5 = 200 km

34. 5 ÷ 20 = 0.25 hours = 15 minutes

35. Speed = 40+10 = 50 km/h; 150 ÷ 50 = 3 hours

36. 60÷30=2 hours, plus 80÷40=2 hours, total 4 hours

37. Combined speed = 40+60 = 100 km/h; 200 ÷ 100 = 2 hours

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38. A, B, C and E (240÷60=4 hours; at 80 km/h, 240÷80=3 hours; in 2 hours the car covers 60×2=120 km, exactly half of 240; at 40 km/h, 240÷40=6 hours; but 20 km every 30 minutes would mean only 40 km/h, not 60 km/h, so D is false)

39. 30÷15=2 hours there, 30÷10=3 hours back, total 5 hours

40. B — Driving time = 120÷60=2 hours; plus 30-minute stop = 2 hours 30 minutes; 08:00 + 2h30 = 10:30

41. First part: 40÷20=2 hours. Total distance = 40+60=100 km; total time = 2+2=4 hours; average speed = 100÷4=25 km/h

42. First part: 12÷6=2 hours. Second part: 6÷12=0.5 hours. Total = 2.5 hours

43. C — Gap grows by 70−50=20 km/h; 20×2=40 km

44. First part takes 200÷50=4 hours; remaining time = 7−4=3 hours; 150÷3=50 km/h

45. Leg times: 60÷60=1 hour, 90÷45=2 hours, total 3 hours. Total distance = 60+90=150 km. Average speed = 150÷3=50 km/h

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